

Project-8: Randomized Algorithms and Complexity

Team Name: AlgoRhythms

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Description:

This project aims to explore and analyze randomized algorithms — algorithms that utilize random numbers to influence their execution path or decisions. Randomized algorithms are widely used due to their simplicity, robustness, and efficiency in solving complex problems where deterministic algorithms may be slower or more cumbersome.

We will implement and study several fundamental randomized algorithms to understand their expected runtime behavior, correctness probability, and how randomness affects their complexity and performance. The results will be analyzed empirically through experiments and statistical measurements.

Algorithms to Analyze:

1. Randomized QuickSort:

- Unlike the deterministic QuickSort, the pivot is randomly chosen from the array.
- This is better because the random pivot selection helps prevent consistently slow performance on certain input patterns, like already-sorted arrays, which can make deterministic QuickSort very slow.
- Randomized QuickSort greatly reduces the probability of the time complexity being $O(n^2)$.
- We analyze randomized QuickSort by estimating average-level efficiency, number of comparisons and robustness against the worst case-inputs due to random pivot selection.

2. Monte-Carlo Primality Test (Miller-Rabin Test):

- The Miller–Rabin primality test is a probabilistic (Monte Carlo) test that checks if a number n is prime.
- It expresses $n - 1$ as $2^k * d$, where d is an odd number.
- The test repeatedly squares $a^d \bmod n$ to detect composites that might pass simpler tests like Fermat.
- If a base fails the test, n is definitely composite. If it passes, n is probably prime. Repeating with multiple bases reduces the probability of error.
- We would analyze Miller-Rabin by evaluating its accuracy, error probability, and run time efficiency with varying number of random bases.

3. Fermat Primality Test:

- The Fermat primality test is a simple probabilistic test based on Fermat's Little Theorem.
- It checks whether $a^{n-1} \equiv 1 \pmod{n}$ for a randomly chosen base a .
- If the congruence fails, n is definitely composite.
- If the congruence passes, n is probably prime.
- The test can be fooled by Carmichael numbers, making it less reliable than the Miller–Rabin test.
- We would analyse Fermat test by calculating its accuracy, error probability, and susceptibility to Carmichael Numbers.
- We would also compare the Fermat and Miller-Rabin tests by finding their respective accuracies, reliabilities and resistance to tricky composites.

4. Karger's Randomized Min-Cut Algorithm:

Cut of a graph: A partition of the vertices of a graph into two disjoint sets. The cut set is the collection of all edges that connect a vertex in one set to a vertex in the other.

- Min Cut of the graph is defined as the cut with the least cut set.
- Karger's Randomized Min-Cut Algorithm is a randomized algorithm to find the min cut.
- This algorithm randomly selects an edge and contracts it. This preserves the cut probability and hence repeating the process increases the probability.
- Expected Complexity: $O(n^2 \log n)$ with a high probability after repetitions.
- We would analyze the probability of correctness as affected by the number of iterations.

5. Monte-Carlo Integration:

- Monte Carlo Integration is a numerical method that estimates definite integrals using random sampling.
- It is particularly useful for high-dimensional or complex integrals where analytical solutions are difficult.
- The algorithm generates random points uniformly over the integration domain. It evaluates the function at these random points and computes the average value.
- This average is then scaled by the volume of the domain to approximate the integral. Monte Carlo Integration converges probabilistically. The accuracy improves with more samples.
- The method can handle irregular domains and multidimensional integrals efficiently.
- The project aims to implement, analyse, and compare the efficiency and accuracy of this algorithm.
- We will analyze Monte Carlo Integration by measuring accuracy versus number of random samples and comparing computed results with exact or benchmark integrals.
- Additionally, you can evaluate time efficiency and convergence behavior as the sample size increases.